

The BUITEMS Undergraduate Prospectus 2025 for the Faculty of Engineering and Architecture (FOE) provides contact information including the UAN 111-717-111 extension 216, the website www.buitms.edu.pk, and the email admissions.faq@buitms.edu.pk. The university holds ISO 9001:2015 TUV Certification, ASQ, and membership in The Association of Commonwealth Universities, operating under the tagline Quality and Excellence in Education. A disclaimer notes that every effort has been made to ensure the information in this Undergraduate Prospectus 2022-23 is correct at the time of printing, but the Faculty of Engineering and Architecture BUITEMS reserves the right to make changes wherever and whenever necessary. In the event of inconsistency between the information contained in this prospectus and the BUITEMS regulations or programs, or where an interpretation of the prospectus is required, the decision of BUITEMS shall be final. The prospectus does not form any part of a contract between any person and the Institute. BUITEMS has given a new vision and a distinct meaning to education, with guidelines for academic achievements that are comparable to those of the most renowned institutions of the world. The university vision is committed to provide quality education with a focus on research and to equip students with the art of living as productive members of the society, contributing to the socio-economic uplift of Pakistan in general, and Balochistan in particular.

Standing as the prestigious and leading university centered in Balochistan influencing the future with academic excellence, BUITEMS is defined by its quality and rigour in education and applied research. BUITEMS has five faculties and each faculty is comprised of many departments, while the Faculty of Engineering and Architecture has eight departments. The University through the offices of the Dean or HODs, Registrar, Controller of Examinations and the Directorates helps in maintaining a conducive environment for teaching and learning, working as a team alongside faculty and students. BUITEMS transformative and innovative education system provides a wide range of academic programs spanning comprehensive areas of engineering, sciences and arts, providing multi-disciplinary and cross-faculty learning opportunities. BUITEMS offers a diverse and comprehensive choice of courses in five faculties with a range of undergraduate and graduate programs, motivating students to explore their interests, discover their talents and pursue their passions for creating and discovering their own BUITEMS experience. The BUITEMS quest for academic excellence is based on five fundamental pillars which are the students, the faculty, the curriculum, the learning environment and the civil society we serve. The university stands committed to follow all statutory requirements in academic offerings not only in following the standards but also to set new Records of Excellence. BUITEMS provides great learning opportunities for students with considerably personalized teaching methods and effective supervisions. As a research-intensive institution with a commitment to provide high-quality education, BUITEMS ensures an academically rigorous and stimulating experience for its students through highly-qualified faculty members, equipped labs and well-structured programs that provide a great learning experience in the undergraduate and graduate degree programs. The faculties responsibilities include designing the curricula, organizing lectures, seminars, practicals, and projects, arranging examinations and marking papers, maintaining academic excellence and professionalism, and providing an environment conducive to learning, teaching, academic inquiry, and innovation.

BUITEMS expects candidates to be individuals who are willing to contribute in the field of science, engineering and arts, strongly encouraging all applicants to choose fields of study which truly interest them. The university is looking for students who have a strong academic potential, have a strong interest in the fields they are applying for, and will benefit from and grow in the BUITEMS environment. Academic potential should be reflected in past

qualifications and the entry test, and BUITEMS expects its students to excel further in terms of academic performance by aiming at an excellent cumulative grade point average throughout the degree program. At BUITEMS, the administration is looking for students who are truly excited about their chosen field of study and want to make that field their career path, firmly believing that academic excellence can only be achieved when the student is motivated and learning comes through stimulation and interest. Therefore, students are strongly advised to think carefully when choosing their field of study. The university is also looking for self-motivated students who can think critically, understand the model of higher learning requiring independent study alongside classroom learning, possess self-discipline, and have a desire to learn.

Applying to BUITEMS involves four simple steps. First, applicants must choose a field or major they will be studying for several years, making sure it interests and excites them. Second, applicants must fill out the admission form available online at buitems.edu.pk/Admission or at the campus admissions office, and submit it online or in person along with the requisite documents and the bank receipt of the admission processing fee. Third, eligible students are invited to take the NTS standard admissions test for a given major. Fourth, admission will be offered based on a merit calculated from the cumulative weight of twenty percent for the marks scored in the Matriculation examination, fifty percent for the marks scored in the Intermediate examination, and thirty percent for the marks scored in the NTS admissions test. Applicants can apply through the online BUITEMS admissions system by downloading and printing the application form, attaching all relevant documents and the application processing fee receipt, and sending the complete hard copy application to the BUITEMS admissions office before the closing date. Entry requirements for undergraduate and graduate programs are available in the Disciplines and Departments section of the prospectus. Supporting documents required include an attested copy of the Secondary School Certificate, an attested copy of the Higher Secondary School Certificate, an attested copy of the Applicant's CNIC or B-form, an attested copy of Local or Domicile, a bank draft or pay order or receipt of online application processing fee, six recent passport size photographs, an attested copy of the CNIC of the applicant's father or guardian, and an attested copy of the Character Certificate from the last institute attended. The non-refundable application processing fee is 3500 Rupees for national applicants and 75 US Dollars for foreign applicants, which must be deposited on or before the last date of application submission. Payment options include a bank draft or pay order drawn in favor of Registrar BUITEMS Quetta, or cash deposited in Account Number 23587000000201 at Habib Bank Ltd BUITEMS University branch in Quetta, Pakistan. Only candidates with complete applications will be issued an admit card for the admission test, and those without it will not be allowed to take the test or interview. The document mailing address is the Admissions Office at BUITEMS Takatu Campus on Airport Road in Baleli, Quetta. Contact numbers include the UAN +92 81 111-717-111, other lines +92 81 2880560 and 2880136, with extensions 163, 216, and 217.

International students are highly encouraged to apply for admission because BUITEMS believes that diversity brings innovation, and many international students are already enrolled in various programs. The eligibility requirements for international applicants are the same as for domestic applicants. An applicant for transfer from a local or foreign institution is required to have passed the BUITEMS admission test by securing equal or more marks than the minimum merit of the program in which the student seeks admission, but acceptance depends on the availability of seats and the quality of academic work already done. As a BUITEMS student in a community of more than 10,000 students and faculty from the world's

leading institutions, you will be part of one of the nation's largest engineering, sciences, and art schools, and you are advised to look up full details on individual fields of study, credit hours, and research aspects to maximize your experience. Regarding frequently asked questions, the duration of BS programs is four years, equally divided into eight semesters, except for the B. Arch program which is spread over five years and ten semesters. Each semester is 18 weeks long, consisting of 16 weeks for teaching and two weeks for midterm and final examinations. The duration for completion of the MS program is 2 years from the date of enrollment. Change of academic program is generally not encouraged but may be allowed on the recommendation of the Chairpersons concerned and approval of the Dean within the first two weeks of the first semester, subject to eligibility, seat availability, and the student's merit being above the merit of the last student admitted into the desired program. Freezing is not allowed in the first semester of a program, but after that, a student can apply to freeze at most two semesters in their entire program of study. A student is required to take 15 to 18 credit hours of course work per regular semester, but the Chairperson may allow a maximum of 21 credit hours with the Dean's approval. Whenever a student's CGPA falls between 1.0 and 2.0, they are put on first probation for the next semester. If they fail to raise the CGPA to 2.0 or above, they are placed on second probation, and if they fail again, they are placed on last probation. If the student fails to raise the CGPA to 2.0 or above in the last probation, they are dropped from the university rolls. Passing requires a minimum of 2.0 CGPA for undergraduate programs and 2.5 CGPA for MS programs. Financial assistance options under the BUIITEMS Fee Concession and Scholarship Policy include merit scholarships available after the first semester, work and study programs, fee concessions for needy students, fee concessions for siblings, fee concessions for dependents of BUIITEMS employees, fee payments in installments, and USAID need-based scholarships, with more information available at the University Advancement and Financial Assistance Office.

Student life at BUIITEMS is a work hard and have fun culture where students, faculty, and staff enjoy an exciting, vibrant, and colorful life featuring high-class academic lectures, laboratory experiments, sports, art events, and service projects. Student affairs comprises more than twenty student clubs and organizations that play a leading role in life at BUIITEMS by organizing cultural festivals and accomplished student performances in theatre, arts, and culture. The annual Olympiad marks the culmination of athletics and sports, providing a week of non-stop competitions in more than fifteen sports categories with over one thousand athletes participating from among the staff and students. The BUIITEMS family has a special motivation for community service, with students and staff contributing to elementary education for child laborers under the Free Citizen Schools organization. This continuous buzz enhances the ability of students and staff to reach their fullest potential through diverse academic, personal, and professional development experiences in a welcoming and stimulating campus life environment. BUIITEMS has more than 3,000 members of its sports facilities and approximately 1,000 students participate in different sports tournaments inside and outside the province. The university offers sports bursaries, funds for team participation in inter-university championships, and provides facilities like a multi-purpose sports complex for basketball, handball, badminton, table tennis, and volleyball, alongside a high-class gym with separate facilities for females, a swimming pool, a bowling club, a PCB-standard cricket stadium, a lush green football ground, and basketball courts.

BUIITEMS spends millions on equipment and academic support services, providing libraries with over 40,000 books, free access to digital resources, and a state-of-the-art automated library management system. The central library system has one central library on each campus and several small departmental libraries. The research center in the Sir Syed Block

provides dedicated research desks equipped with high-performance computers for graduate students, while senior design labs cater to undergraduate students working on final-year projects. The campus also has art galleries supported by the Faculty of Arts and Basic Sciences showcasing modern art, artifacts, paintings, and sculptures. Furthermore, the National Incubation Center Quetta supports the youth of Balochistan to develop sustainable startup ventures, create jobs, commercialize new technology, and strengthen the local economy. Quetta is an easily accessible high-altitude city situated at 1,680 meters above sea level, famous for its amazing weather, beautiful fruit orchards, and semi-arid climate. The winter season starts in late November and ends in late March with snow as the main precipitation and temperatures around 4 to 5 degrees Celsius. Spring is from early April to late May, summer is from late May to early September with temperatures ranging from 24 to 26 degrees Celsius, and autumn is from late September to mid-November with temperatures between 12 and 18 degrees Celsius. Attractions near the city include Ziarat, which is 130 kilometers away and features the second-largest Juniper forest in the world alongside the historic Ziarat Residency where Quaid-e-Azam Muhammad Ali Jinnah spent his last days. Other attractions include Hanna Lake with its turquoise water and lakeside restaurant 10 kilometers east of the city, and Kharkhasa located 10 kilometers west of Quetta in a 16 kilometer long narrow valley featuring the Chiltan Hill Viewpoint. Nearby cities of Kirani and Harnai are also popular scenic locations. Quetta features museums such as the Quetta Geological Museum on Sariab Road, the Command and Staff College Museum dedicated to British military history, and the Quetta Archaeological Museum on Wafa Road showcasing artifacts from Mehrgarh. The Wali Tangi Dam is a small dam in the Urak Valley that stores fresh water from the Zarghoon Hills, and the Hazarganji Chiltan National Park is a protected mountainous park located 20 kilometers southwest of Quetta. Finally, Pir Ghaib is a waterfall and picnic spot located 70 kilometers from Quetta City in the historic Bolan Valley.

The Faculty of Engineering and Architecture was established in 2004 at the BUITEMS Takatu Campus, and its engineering education focuses on exploiting natural resources by producing well-trained engineers for industries like oil and gas, mining, dam building, construction, petrochemicals, and fertilizers. The faculty initially opened with the Petroleum and Gas Engineering, Textile Engineering, and Chemical Engineering departments, but since 2007, six more departments were established including Civil Engineering, Geological Engineering, Mechanical Engineering, and Architecture, all offering BS, MS, and PhD degree programs. All BS Engineering programs are accredited by the Pakistan Engineering Council, and the Architecture program is accredited by the Pakistan Council for Architecture and Town Planning. In his message, Dean Prof. Dr. Ghulamullah Khan states that the faculty is a place where innovation meets excellence, committed to providing a rigorous academic environment that combines strong theoretical foundations with cutting-edge research and real-world practice, while instilling ethical values, professional integrity, and social responsibility in students. BUITEMS follows Outcome-Based Education, which is an educational theory that structures each part of an instructional system around goals or outcomes so that by the end of the educational experience, each student should achieve the specified goals. Outcome-Based Education utilizes student learning outcomes to make explicit what a student is expected to know, provides positive learning activities to help students attain Course Learning Outcomes, and assesses student performance through explicit criteria. In 1989, the Washington Accord established an international accreditation covenant for professional engineering undergraduate programs, recognizing the structural equivalency of accredited programs across member countries. The Pakistan Engineering Council is a signatory and implements Outcome-Based Education across all engineering degree-awarding institutes in Pakistan, meaning BUITEMS aligns all engineering programs with the guidelines

of the Outcome-Based Education manual. Key features of this system include utilizing Program Educational Objectives, Course Learning Outcomes, and Performance Indicators. It is objective and outcome-driven, centered around the requirements of students and key stakeholders, with Program Educational Objectives addressing graduate achievements 3 to 5 years post-graduation focusing on Knowledge, Skills, and Attitudes. Program Learning Outcomes map directly to Program Educational Objectives, and Course Learning Outcomes map directly to Program Learning Outcomes. The 12 Graduate Attributes or Program Learning Outcomes include Engineering Knowledge to solve complex engineering problems, Problem Analysis to research and analyze complex problems, and Design and Development of Solutions to meet public health, safety, societal, cultural, and environmental needs. They also include Investigation of complex problems methodically, Modern Tool Usage to apply appropriate techniques and IT tools, and The Engineer and Society to apply contextual reasoning to professional engineering practice. Further attributes are Environment and Sustainability to understand the impact of engineering solutions, Ethics to apply ethical principles and professional responsibilities, Individual and Team Work in diverse settings, Communication skills to report and present complex activities, Project Management to apply engineering principles in multidisciplinary environments, and Lifelong Learning to recognize the importance of continuous technological innovation.

The Department of Architecture addresses urbanization challenges in Pakistan and the need for sustainable built environments and infrastructure in Balochistan, training architects to design new buildings, restore old structures, and develop uses for existing buildings. Founded in September 2009 with 4 faculty members and 40 students, it now has 19 faculty members and offers a 5-year Bachelor of Architecture degree. Its vision is to be among the best architecture schools where professionals explore and build culture artistically with creativity, while its mission is committed to training students through in-depth study in social, cultural, political, and historic contexts for comprehensive design solutions. The Program Educational Objectives focus on introducing architectural fundamentals, practicing theoretical and practical foundations, understanding elements of design, training students in integrated design of built and un-built environments, and innovating within local and regional contexts. The curriculum spans 174 credit hours and 66 courses, requiring prerequisites of F.Sc. Pre-Engineering or equivalent with at least 50% marks, or a Diploma with at least 60% marks, alongside the BUIITEMS admission test and an aptitude test covering creative thinking and free-hand drawing. The scheme of study covers ten semesters including courses like Foundation Studio, Visual Communication Skills, History of Architecture, Architectural Studio, Building Services and Systems, Urban Planning and Design, Landscape Architecture, Heritage Conservation, and Thesis Design. The faculty main team includes Chairman Dr. Ar. Waqas Ahmed Mahar with a PhD from the University of Liège, Dr. Sohrab Ahmed Marri, Ar. Palwasha Amanullah, Ar. Muhammad Sohail, Ar. Azra Alvi, Ar. Ariana Bazai, Ar. Mamoon ur Rashid, Ar. Umer Ahmed, Ar. Jahangeer Khan, Ar. Jalal Faisal, Ar. Naveed Ur Rehman Anwar, Ar. Tahira Shaukat, Ar. Adeel Zahoor, Ar. Gullaly Javed, Ar. Maskha Khan, Ar. Shehroze Shah, Ar. Sabeen Mehmood Durrani, Ar. Babar Khan, Ar. Mir Behram Khan Raisani, Ar. Asma Batool, Ar. Syed Abdullah Sami, Ar. Sanaullah Aziz, Ar. Urooj Fatima Nagi, Ar. Ali Gul Khan, and Ar. Zakir Hassan.

The Department of Chemical Engineering applies physical and life sciences with mathematics for mineral-based industries, petrochemical plants, pharmaceuticals, synthetic fibers, and refineries. Founded in 2005 with 4 faculty members and 15 students, it now has 21 faculty members and offers BS and MS degrees. Core research areas include nanotechnology, algal cultivation, biomass and bioenergy, waste management, wastewater treatment, and

polymer composites. Its vision is to be a leading department providing quality graduates, and its mission establishes a conducive environment for professional education and research. The Program Educational Objectives aim for professional competency to address industry demands, contributing to societal problems, and pursuing continuous technical achievements. The curriculum comprises 136 credit hours and 48 courses, requiring F.Sc. Pre-Engineering or a Diploma of Associate Engineer with at least 60% marks and passing the BUITEMS admission test. Assessment follows Outcome-Based Education guidelines where lab work is assessed via Lab Rubrics and Program Educational Objectives are evaluated three years post-graduation. The eight-semester study scheme includes subjects like Organic and Inorganic Chemistry, Chemical Process Calculations, Fluid Mechanics, Chemical Engineering Thermodynamics, Separation Processes, Heat Transfer, Chemical Reaction Engineering, Chemical Plant Design, Instrumentation and Process Control, and Industrial Management. The main faculty team consists of Chairman Dr. Muhammad Amin, Dr. Muhammad Najam Khan, Dr. Asadullah Baloch, Dr. Faisal Mushtaq, Dr. Ghulam Ullah Kakar, Dr. Syed Kamran Sami, Dr. Syed Haseeb Sultan, Engr. Zahid Naeem Qaisrani, Engr. Mohammad Siddique, Dr. Abdul Qadeer Dayo, Engr. Awais Sattar Ghouri, Engr. Azmat Ali Khan, Engr. Imran Khan, Engr. Tufail Mustafa, Engr. Iqra Anjum, Engr. Nawab Zada, Engr. Saadat Ullah Khan Suri, Engr. Abeera Malik, Engr. Muhammad Farooq Sajjad, Engr. Hafiz Muhammad Babar Munir, and Engr. Gohram Khan.

The Department of Civil Engineering covers planning, design, budgeting, surveying, construction, and maintenance of physical structures like roads, bridges, canals, dams, airports, and railways. Established in 2008 with an initial intake of 100 students, it offers BS and MS in Civil Engineering. Its vision is to be a Centre of Excellence in education and research on structures, environment, and highways, and its mission is to produce highly competent professionals of high ethical values. The Program Educational Objectives ensure graduates gain civil engineering skills, develop expertise through lifelong learning, pursue research and higher studies, and work as responsible professionals independently or in teams. The program requires 136 credit hours and 47 courses, taking students with F.Sc. Pre-Engineering or DAE with at least 60% marks who qualify the BUITEMS admission test. It adheres to the Pakistan Engineering Council Outcome-Based Education guidelines. Over eight semesters, courses include Engineering Surveying, Mechanics of Solids, Construction Engineering, Structural Analysis, Soil Mechanics, Concrete Technology, Engineering Hydrology, Highway and Traffic Engineering, Reinforced Concrete Design, Geotechnical and Foundation Engineering, Geoinformatics, Environmental Engineering, and Steel Structures. The main faculty team features Chairman Dr. Muhammad Habib, Associate Dean Dr. Zafar, Dr. Naik Muhammad Babar, Dr. Saeed ullah Jan Mandokhel, Dr. Asad Naeem, Engr. Nawaz Ali, Engr. Abdullah Khan, Engr. Syed Zafarullah Khan, Engr. Shahnawaz, Engr. Meer Hamza, Engr. Amira Raffique Ahmed, Engr. Mohammad Iqbal Tareen, Engr. Syed Muhammad Qaisar Farooq Shah, Engr. Marjan Gul, Engr. Zarak Khan, Engr. Shehzada Muhammad Emaad Khan, Engr. Muhammad Junaid Jamil, Engr. Arian Khan, Engr. Samiullah Khan, Engr. Muhib Ullah Tareen, Engr. Aamir Mahmood, Dr. Mohammad Salman Nazar, Engr. Syed Jahanzaib Shah, Dr. Syed Nasir Shah, Engr. Mohammad Ayoub, Engr. Anwaar Hazoor Ansari, Engr. Zafreen Elahi, Engr. Ahsan Ali, Engr. Zarak Khan Kasi, Engr. Syed Abdul Ghaffar, Engr. Abdul Wakeel Khan, Engr. Ali Fahad Khan, Engr. Masood Ahmed, and Engr. Muhammad Irfan Ullah.

The Department of Geological Engineering applies engineering principles to geological materials for infrastructure design and resources exploration. Founded in 2008 with an initial batch of 33 students, it now has 14 faculty members and offers a BS in Geological

Engineering. Its vision is to be a leading department providing quality graduates, and its mission is committed to training engineers with advanced knowledge and promoting professional competence. The Program Educational Objectives focus on applying technological fundamentals to geotechnical issues, exhibiting effective communication and leadership skills, and fulfilling ethical and social responsibilities. The 136 credit hour curriculum of 46 courses requires F.Sc. Pre-Engineering, Pre-Medical, ICS, or a relevant DAE with at least 60% marks and passing the BUITEMS admission test, with mandatory non-credit courses for ICS and Pre-medical students in the first semester. The eight semesters cover Physical Geology, Mineralogy and Petrology, Stratigraphy and Structural Geology, Hydrogeology, Engineering Geology, Geotechnical Engineering, Explosives Engineering, Petroleum Geology, Rock Mechanics, Earthquake Seismology, GIS and Remote Sensing, Tunneling and Excavation Engineering, and Drilling Engineering. The main faculty includes Chairman Dr. Shabir Ahmed, Dr. Abdul Muntaqim Naji, Mr. Shaukat Parvaiz, Engr. Imad Ali, Mr. Nasir Khan, Engr. Siraj Hassan, Engr. Muhammad Hasnain, Engr. Rafi Ullah, Engr. Barkat ullah Khan, Engr. Asif Abbas, Engr. Saleem Ullah, and Engr. Rana Muhammad Faisal Khan.

The Department of Mechanical Engineering deals with processes ranging from small components to large plants operating across manufacturing, power, medical, aerospace, oil, and resource management fields. Founded in 2014 with 3 faculty members and 50 students, it now has 11 faculty members and offers a BS in Mechanical Engineering. Its vision is to offer a unique blend of specializations, and its mission provides a challenge-driven academic environment to build student capacity. The Program Educational Objectives involve solving energy and manufacturing sector problems, designing mechanical systems with modern techniques, addressing sustainability with managerial skills, and cultivating lifelong learning. The 134 credit hours program of 48 courses requires F.Sc. Pre-Engineering or a relevant DAE with 60% marks and follows the Outcome-Based Education framework. The eight-semester syllabus includes Engineering Materials and Metallurgy, Engineering Statics, Thermodynamics, Engineering Dynamics, Solid Mechanics, Fluid Mechanics, Machine Design, Heat and Mass Transfer, Manufacturing Processes, Mechanics of Machines, Mechanical Instrumentation and Control, Internal Combustion Engines, Mechanical Vibrations, Refrigeration and Air Conditioning, Automation and Robotics, and Renewable Energy Systems. The main faculty team consists of Chairman Dr. Azmatullah Khan Sherani, Dr. Ali Nawaz Mengal, Engr. Arshad Ali, Dr. Asadullah, Dr. Saddam Hussain Soomro, Engr. Asif Hussain, Engr. Muhammad Adil Khan, Engr. Kaleemullah Shaikh, Engr. Muhammad Ayub, Engr. Zaib Shahid, Engr. Arsalan Yawar, Engr. Shabir Ahmed, Engr. Gohar Ali, Engr. Qamar ud Din Abid, Engr. Moiz Ullah, and Engr. Suleman khan.

The Department of Mining Engineering focuses on the discovery, evaluation, exploitation, development, and processing of minerals. Founded in 2008 with 4 faculty members and 22 students, it now has 16 faculty members and offers BS and MS degrees. Its mission is to produce technically competent, environmentally friendly mining graduates through an advanced curriculum. The Program Educational Objectives aim to address issues related to the exploitation of mineral deposits, exhibit high-level technical competency, exercise professional excellence focusing on workplace safety, and persuade lifelong learning models. The curriculum has 140 credit hours and 43 courses, requiring F.Sc. Pre-Engineering or a relevant DAE with at least 60% marks. Structured under the Pakistan Engineering Council Outcome-Based Education manuals, the study scheme over eight semesters includes Applied Geology, Mine Surveying, Structural Geology, Mineral Exploration, Explosives Engineering, Cement Technology, Rock Mechanics, Coal Technology, Drilling Technology, Mine

Machinery, Mine Ventilation, Mineral Processing, Underground Mine Design, Tunnel Engineering, and Mining Laws. The faculty team comprises Chairman Dr. Hafeezur Rehman, Dr. Fahim Ahmed Ibupoto, Engr. Zaheer Ahmed, Dr. Muhammad Bilal, Dr. Wahid Ali, Engr. Muhammad Ali, Engr. Fawad Ul Hassan, Engr. Sher Bacha, Engr. Abdullah Rasheed Qureshi, Engr. Naseer Muhammad Khan, Engr. Tufail Ahmed, and Engr. Abbas Hussain.

The Department of Petroleum and Gas Engineering concerns the maximum sustainable extraction of hydrocarbon elements from deep-surface reservoirs, requiring deep mastery over drilling, logistics, production, and structural reservoir simulation. Launched in 2004 with 2 faculty members and 50 students, it now has 14 faculty members and offers a BS in Petroleum and Gas Engineering. Its vision is to be a Centre of Excellence producing technically sound petroleum engineers, and its mission cultivates industrious engineers by providing advanced technical knowledge. The Program Educational Objectives address complex engineering problems of conventional and unconventional reservoirs, fulfill necessary skills for higher studies, and adopt an environment focused on moral values and lifelong learning. The 136 credit hour program of 44 courses requires F.Sc. Pre-Engineering or a relevant DAE with 60% marks and is aligned with Outcome-Based Education instructional systems. The eight semesters cover Fundamentals of Petroleum Engineering, Petroleum Geology and Geophysical Exploration, Mechanics of Materials, Drilling Engineering, Petrophysics, Properties of Reservoir Fluids, Well Logging, Reservoir Engineering, Petroleum Production Engineering, Petroleum Refinery Engineering, Well Testing, Reservoir Simulation, Natural Gas Engineering, and Principles of Enhanced Oil Recovery. The main faculty team features Chairman Dr. Azizullah Shaikh, Engr. Zafarullah, Engr. Imtiaz Ali, Engr. Allah Bakhsh, Engr. Muhammad Yaqoob Khan, Engr. Shahpur Azam, Engr. Syed Jamal ud din, Engr. Nouman Zobby, Engr. Zeeshan Khan, Engr. Rehan Mansoor, Engr. Najeeb Ullah Khan, Mr. Muhammad Yasir, Engr. Usama Ahmed Khan, and Engr. Saud Ahmed Shah.

The Department of Textile Engineering and Design focuses on the structural configurations between raw materials, processing lines, and physical fabric dynamics to secure desired aesthetic or industrial attributes. Founded in 2004 with 3 faculty members and 20 students, it now has 11 engineering faculty members and offers BS and MS degrees in Textile Engineering and a BS in Fashion and Textile Design. Its vision is to be the leading textile department for Center of Excellence, and its mission is to provide professional training to create sustainable solutions preserving the environment. The Program Educational Objectives aim to implement real-world applications utilizing engineering knowledge, generate innovative sustainable solutions for the textile industry, and advance careers through leadership strategy and understanding of socio-economic and environmental challenges. The 136 credit hour textile engineering program requires F.Sc. Pre-Engineering, Pre-Medical, ICS, or DAE in Textile Engineering with at least 60% marks. The eight-semester study scheme includes Textile Raw Materials, Fibre Science, Spinning Preparatory Process, Fabric Manufacturing, Yarn Production Process, Weave Designs and Calculation, Pre-treatments in Textile Wet Processing, Garments Manufacturing Fundamentals, Textile Dyes and Dyeing, Technical Textiles, Mechanism of Garment Machines, Denim Technology, Textile Testing and Quality Control, and Textile Printing and Finishing. The main faculty team includes Chairman Dr. Muhammad Qasim Siddiqui, Dr. Syed Zameer ul Hassan, Engr. Muhammad Ali Zeeshan, Engr. Muhammad Kashif, Dr. Ali Asghar, Dr. A. Malik Rehan Abbasi, Dr. Ahmer Hussain Shah, Engr. Nazakat Ali Khoso, Engr. Syed Qutaba Bin Tariq, Dr. Saif Ullah Khan, Engr. Lubna Syed, Engr. Qurat-ul-ain Arif, Engr. Raazia Ismat, Engr. Ali Raza

Shafqat, Ms. Maliha Khan Tareen, Ms. Afshan Ambreen, Ms. Elaha Ahmed, and Ms. Misbah Hamad.

The BS Fashion and Textile Design degree maps out hands-on sensitivity for innovative draping, custom weaving profiles, graphic lines, pattern modeling, and advanced fabric design. Offered since Fall 2018, the program won first prize in the All Pakistan DICE Textile Design Competition and third position in a European Union Facemask Competition, aiming to create highly skilled designers and entrepreneurs tailored to both local and international markets. The curriculum consists of 135 credit hours and 60 courses, requiring an Intermediate or equivalent with at least 45% marks, and admissions are awarded based on NTS, aptitude tests, freehand drawing, and a descriptive interview. The eight semesters cover subjects like Geometry and Drafting, Pattern Drafting, Hand Sewing and Embellishment, Yarn Making and Design, History of Costume, Textile CAD, Weave Structure and Design, Pattern Making, Sewing, Draping, Fashion Design, Knitwear Construction Design, Clothing Productivity and Quality Management, Digital Design, Apparel Costing, Denim Product Development, and Studio Execution.

Quetta is the capital city of Balochistan and sits at an average altitude of 1,680 meters, easily accessible by modern transportation infrastructure. Road distances to Quetta include 698 kilometers from Karachi via National Highway N25, 622 kilometers from Multan, 827 kilometers from Peshawar via N50 and N55, 960 kilometers from Lahore, 911 kilometers from Islamabad via N50, and 1373 kilometers from Gilgit. For transit pathways, BUITEMS is located 3 kilometers away from Quetta International Airport, and the Quetta Railway Station links to Karachi via an 863 kilometer track, Lahore, and Peshawar via extensive networks established since the 1890s. The campus map legend denotes locations including the Main Entrance to the Campus, HBL BUITEMS Branch, Cafeteria, Blue Chip Lab, Academic and Admission Admin Block, Entrance Gate to the Campus, Sir Syed Block A and B, Hall One, Library, Pro VC Secretariat, Tuck Shops, CAD CAM Research Hall, JICA Hall, AG Hall, Parking, Textile Labs and Studios, Arfa Karim Randhawa Expo Centre, Orange Hall, Pink Hall, Purple Hall, Green Hall, National Incubation Centre Quetta, Directorate of UA and FA, Jinnah Hall, Iqbal Hall, Sports Complex, Markhor Auditorium, Mosque, VC Secretariat, Cricket Ground, Football Ground, and the Medical Unit.